# DEEPAKKUMAR S

*Computer Science Graduate | Python, SQL, REST APIs, Machine Learning & Generative AI*

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## PROFESSIONAL SUMMARY

Computer Science graduate (8.72 CGPA) specializing in Data Science with hands-on proficiency in Python, SQL, REST APIs, and Machine Learning pipelines. Demonstrated capability in engineering full-cycle data-driven solutions, developing modular backend services, and deploying offline-capable NLP architectures. Proven focus on clean software architecture, database query optimization, and localized data privacy.

## TECHNICAL SKILLS

**Languages & Development:** Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, REST APIs, OOP Concepts

**Databases & Backend:** MySQL, SQLite, Relational DBMS Concepts, CRUD API Design

**AI & Machine Learning:** NLP Basics, Scikit-learn, Time-Series Forecasting (ARIMA, SARIMAX, Prophet, LSTM), Generative AI Tools (Gemini, Claude)

**Data Analytics:** Data Cleaning, Data Visualization, ETL Concepts

**Tools & Practices:** Git, GitHub, Tableau, Power BI, Excel, Jupyter Notebook Agile Basics, Debugging

## PROJECT EXPERIENCE

**MILO — Offline AI Personal Assistant** | *Python, Whisper STT, SQLite, NLP*

- Engineered a 100% offline, privacy-first personal assistant in Python using Whisper STT and NLP, eliminating external API latency and cloud dependencies.
- Architected local data storage pipelines using SQLite, streamlining structured data logging for reminders and real-time expense tracking.
- Implemented asynchronous audio-processing pipelines to ensure real-time command execution directly on-device with zero data leaks.

**Event Management API System** | *Node.js, TypeScript, Express, SQLite*

- Developed a production-grade RESTful API backend using Node.js, TypeScript, and Express, implementing normalized SQLite schemas for optimized transactional queries.
- Standardized complete request-response validation contracts by drafting interactive Swagger/OpenAPI documentation, improving endpoint testability.
- Accelerated backend system architecture design and edge-case debugging cycles leveraging Generative AI workflows. .

**Stock Price Prediction System** | *Python, ARIMA, SARIMAX, Prophet, LSTM, Scikit-learn*

- Designed an end-to-end time-series forecasting pipeline in Python, evaluating ARIMA, SARIMAX, Prophet, and LSTM architectures on historical equity data.
- Optimized feature engineering and sliding-window sequences using Pandas and NumPy to benchmark prediction accuracy.
- Achieved an 18% reduction in RMSE with the LSTM model over traditional statistical baselines, significantly minimizing predictive error.

## EDUCATION

- B.E. Computer Science Engineering (Data Science), Annamalai University (2022–2026) - CGPA: 8.72
- 12th Standard, Govt. Higher Secondary School (2021) - 85%
- 10th Standard (2019) - 76%

## CERTIFICATIONS & WORKSHOPS

- Python & SQL - HackerRank, 2026
- Basics of Python - Infosys Springboard, 2026
- Critical Thinking in the AI Era - HP LIFE, 2026
- Employability Enhancement Training - E-Square (Machine Learning, DBMS, Cloud Concepts, R)
- Git & GitHub Workshop - Version Control & Team Collaboration Workflows
- Event Organizer, COMPSEM Poster Designing Event - Coordinated cross-team logistics

## LANGUAGES

English, Tamil